

Reinforcement Learning

Quiz 1

16/09/2025

Sanjit K. Kaul

Instructions: You have 70 minutes to work on the questions. Answers with no supporting steps will receive no credit. No resources, other than a pen/pencil, are allowed. In case you believe that required information has not been provided, make a suitable assumption.

Question 1. 40 marks We have two arms 1 and 2 in a casino. Arm 1 is configured to give a reward of 1 with probability p and 0 otherwise. Arm 2 is configured to give a reward of 1 with probability q and 0 otherwise. Assume $p > q$. A player enters the casino to play the bandit game, which allows a player multiple attempts where in every attempt the player must choose one arm. The player accumulates the rewards given by the arm picks and doesn't know the distributions of rewards of the arms.

Assume a very large number of attempts. Further suppose the player uses the sample mean to update estimates of the expected values of the arms. The initial values of the estimates are set to 0. Suppose the player uses the greedy policy to choose an arm, which picks the arm with the larger estimate. If both arms have the same estimate, the player picks either arm with equal probability. Calculate the probability that in the long run the player will pick the optimal arm. Calculate the expected reward obtained by such a player.

Calculate the expected reward obtained by a player that in every attempt picks either arm with equal probability. How does it compare with the player above who chooses the greedy action?

Calculate the expected reward obtained by a player that uses the ϵ -greedy policy that explores with probability $\epsilon > 0$. How does this compare with the above two policies for different selections of $\epsilon \in (0, 1)$?

Question 2. 15 marks A shopkeeper must decide on the number of milk cartons he must stock at the beginning of every day. Given the space, the shopkeeper can stock at most m milk cartons. Thus the shopkeeper may choose to stock a number of milk cartons in $\{0, 1, \dots, m\}$. Any milk cartons that are not purchased at the end of the day must be discarded and result in a corresponding loss, with every discarded milk carton resulting in a loss of INR 5. Every sold milk carton results in a profit of INR 1. Every customer that returns without being able to purchase a milk carton because none are available on the shelf corresponds to a loss of INR 1. The shopkeeper doesn't know the distribution of customer demand and must optimize the number of milk cartons to be placed on the shelf so that the cumulative rewards at the end of 365 days are maximized.

The shopkeeper approaches you to propose a way. Devise an algorithm that every morning tells the shopkeeper the number of milk cartons he must stock. Provide a pseudo-code that clearly describes the workings of your algorithm. State any assumptions you make.

Question 3. 15 marks Let A be the set of all possible actions and a be any action in A . Let $H(x)$ be a function that assigns a value to every action $x \in A$. Calculate the gradient of $e^{H(a)} / (\sum_{b \in A} e^{H(b)})$ with respect to the vector of preference values $H(a), a \in A$.

Question 4. 15 marks At time t , due an error in its observation system, an agent is unable to observe the state S_t . However, the agents has stored observed states $s_0, s_1, \dots, s_{t-1}$. The agent knows the MDP and also knows

the optimal value function $v_*(s)$ for every state s . The agent would like to choose action a_t such that its expected value of the return G_t is maximized. Derive the optimal action in terms of the MDP function $p(s', r|s, a)$, policy π , if any, as required, and other quantities that are said to be known to the agent.

Question 5. 15 marks You are given two value functions $v_1(s)$ and $v_2(s)$, where state $s \in \{1, 2, 3, 4, 5, 6\}$. Let $v_1(s) = v_*(s)$ for $s = 1, 2, 3$ and let $v_1(s) = v_*(s) - \epsilon s^2$ for $s = 4, 5, 6$. Here $v_*(s)$ is the optimal value and $\epsilon > 0$. Further, let $v_2(s) = v_*(s)$ for $s = 4, 5, 6$ and let $v_2(s) = v_*(s) - \epsilon s^3$ for $s = 1, 2, 3$. Derive the optimal policy function for any MDP $p(s', r|s, a)$.

Question 1. 40 marks We have two arms 1 and 2 in a casino. Arm 1 is configured to give a reward of 1 with probability p and 0 otherwise. Arm 2 is configured to give a reward of 1 with probability q and 0 otherwise. Assume $p > q$. A player enters the casino to play the bandit game, which allows a player multiple attempts where in every attempt the player must choose one arm. The player accumulates the rewards given by the arm picks and doesn't know the distributions of rewards of the arms.

- (a) Assume a very large number of attempts. Further suppose the player uses the sample mean to update estimates of the expected values of the arms. The initial values of the estimates are set to 0. Suppose the player uses the greedy policy to choose an arm, which picks the arm with the larger estimate. If both arms have the same estimate, the player picks either arm with equal probability. Calculate the probability that in the long run the player will pick the optimal arm. Calculate the expected reward obtained by such a player. } $\frac{28}{40}$
- (b) Calculate the expected reward obtained by a player that in every attempt picks either arm with equal probability. How does it compare with the player above who chooses the greedy action? } $\frac{6}{40}$
- (c) Calculate the expected reward obtained by a player that uses the ϵ -greedy policy that explores with probability $\epsilon > 0$. How does this compare with the above two policies for different selections of $\epsilon \in (0, 1)$? } $\frac{6}{40}$

(a) The key observation is that both arms keep getting picked with equal probability till the attempt in which for the very first time a reward of 1 is obtained.

Suppose attempt $L=1$ is the first attempt during which a reward of 1 is received. For attempts $1, 2, \dots, L-1$, the rewards must have been 0.

Further observe that if arm 1 was the arm chosen at attempt L , given the greedy policy & sample mean based estimates, arm 1 will be chosen in attempts $L+1, L+2, \dots$. On the other hand, if arm 2 was the arm chosen at attempt L , arm 2 will be chosen in attempts $L+1, L+2, \dots$.

The optimal arm (which is arm 1 in our problem) will be chosen if, in attempt L , arm 1 is chosen & gives a reward of 1.

$P[\text{Optimal arm is chosen in the long run}]$

$$\begin{aligned} &= \sum_{L=1}^{\infty} P[L=L, \text{arm 1 is chosen in attempt } L \text{ \& gives a reward of 1}] \\ &= \sum_{L=1}^{\infty} \left(\frac{1}{2} (1-p) + \frac{1}{2} (1-q) \right)^{L-1} \cdot \frac{1}{2} p \\ &= \sum_{L=1}^{\infty} \frac{1}{2^L} (2 - (p+q))^{L-1} p \\ &= \frac{p}{2} \sum_{L=1}^{\infty} \left(1 - \frac{(p+q)}{2} \right)^{L-1} \\ &= \frac{p}{2} \left(1 + \left(1 - \frac{(p+q)}{2} \right) + \left(1 - \frac{(p+q)}{2} \right)^2 + \dots \right) \\ &= \frac{p}{2} \left(\frac{1}{1 - \left(1 - \frac{(p+q)}{2} \right)} \right) \\ &= \frac{p}{2} \cdot \frac{2}{p+q} = \frac{p}{p+q}. \end{aligned}$$

Calculation of the probability via any legitimate means.
Up to $\frac{28}{40}$

Expected reward (in the long run)

$$\begin{aligned} &= \left(\text{Expected reward} \mid \text{optimal arm is chosen in the long run} \right) \frac{p}{p+q} \\ &\quad + \left(\text{Expected reward} \mid \text{the other arm is chosen in the long run} \right) \frac{q}{p+q} \\ &= (p) \left(\frac{p}{p+q} \right) + q \left(\frac{q}{p+q} \right) = \frac{p^2 + q^2}{p+q}. \end{aligned}$$

Expected reward Up to $\frac{8}{28}$

(b) If the player picks either arm with equal probability, the expected reward from any attempt is $\frac{1}{2}p + \frac{1}{2}q$.
 $= \frac{p+q}{2}$.

Note that

$$\begin{aligned} \frac{p^2 + q^2}{p+q} &= \frac{(p+q)^2 - 2pq}{p+q} = p+q - \frac{2pq}{p+q} \\ &= \frac{p+q}{2} + \frac{p+q}{2} - \frac{2pq}{p+q} \\ &= \frac{p+q}{2} + \frac{(p+q)^2 - 4pq}{2(p+q)} \\ &= \frac{p+q}{2} + \frac{(p-q)^2}{2(p+q)} > \frac{p+q}{2}. \end{aligned}$$

The player choosing the greedy action is guaranteed a larger expected reward.

Calculation: up to $\frac{3}{6}$

Comparison Up to $\frac{9}{6}$

(c) In the long run, the expected reward using the ϵ -greedy policy is (for $\epsilon > 0$):
 $(1-\epsilon)p + \frac{\epsilon}{2}p + \frac{\epsilon}{2}q$
 $= (1-\frac{\epsilon}{2})p + \frac{\epsilon}{2}q$

Calculation: Up to $\frac{3}{6}$

We get the expected reward from the greedy policy by setting $\frac{\epsilon}{2} = \frac{q}{p+q}$.

We get the expected reward from the policy that chooses arms with equal probability by setting $\epsilon=1$.

For $\frac{\epsilon}{2} < \frac{q}{p+q}$, ϵ -greedy will give an expected reward greater than the greedy policy.

Comparison Up to $\frac{3}{6}$

Question 2. 15 marks A shopkeeper must decide on the number of milk cartons he must stock at the beginning of every day. Given the space, the shopkeeper can stock at most m milk cartons. Thus the shopkeeper may choose to stock a number of milk cartons in $\{0, 1, \dots, m\}$. Any milk cartons that are not purchased at the end of the day must be discarded and result in a corresponding loss, with every discarded milk carton resulting in a loss of INR 5. Every sold milk carton results in a profit of INR 1. Every customer that returns without being able to purchase a milk carton because none are available on the shelf corresponds to a loss of INR 1. The shopkeeper doesn't know the distribution of customer demand and must optimize the number of milk cartons to be placed on the shelf so that the cumulative rewards at the end of 365 days are maximized.

The shopkeeper approaches you to propose a way. Devise an algorithm that every morning tells the shopkeeper the number of milk cartons he must stock. Provide a pseudo-code that clearly describes the workings of your algorithm. State any assumptions you make.

We have $m+1$ arms.

Choice of arm, gives a reward at the end of the day.

$$\rightarrow \text{reward } R_t = (k-u)^+(-5) + \min(k,u)(1) + (u-k)^+(-1),$$

where $k \in \{0, 1, \dots, m\}$ is the choice of arm,
 u is the number of customers that arrive to purchase a milk carton.

Any choice of mean for estimating values $Q(k), k \in \{0, 1, \dots, m\}$ is fine. Sample mean or exponentially weighted mean.

Given a choice of mean & clearly calculated rewards, any bandit algorithm is fine.

} Clearly calculated rewards
 Up to 15

} Up to 15

} Pseudo code. Up to 15

Question 3. 15 marks Let A be the set of all possible actions and a be any action in A . Let $H(x)$ be a function that assigns a value to every action $x \in A$. Calculate the gradient of $e^{H(a)} / (\sum_{b \in A} e^{H(b)})$ with respect to the vector of preference values $H(a), a \in A$.

We need to worry about the following two cases for calculating the gradient vector.

Consider action a :

$$\begin{aligned} & \frac{\partial}{\partial H(a)} \frac{e^{H(a)}}{\sum_{b \neq a} e^{H(b)} + e^{H(a)}} \\ &= \frac{e^{H(a)}}{\sum_b e^{H(b)}} - \frac{e^{H(a)} e^{H(a)}}{\left(\sum_b e^{H(b)}\right)^2} \\ &= \frac{e^{H(a)} \left(\sum_b e^{H(b)} - e^{H(a)}\right)}{\left(\sum_b e^{H(b)}\right)^2} = \frac{e^{H(a)} \sum_{b \neq a} e^{H(b)}}{\left(\sum_b e^{H(b)}\right)^2} \end{aligned}$$

Up to
7.5/15

Consider action $b \neq a$:

$$\begin{aligned} & \frac{\partial}{\partial H(b)} \frac{e^{H(a)}}{\sum_{k \neq b} e^{H(k)} + e^{H(b)}} \\ &= - \frac{e^{H(a)} e^{H(b)}}{\left(\sum_k e^{H(k)}\right)^2} \end{aligned}$$

Up to
7.5/15

Question 4. 15 marks At time t , due an error in its observation system, an agent is unable to observe the state S_t . However, the agents has stored observed states $s_0, s_1, \dots, s_{t-1}$. The agent knows the MDP and also knows the optimal value function $v_*(s)$ for every state s . The agent would like to choose action a_t such that its expected value of the return G_t is maximized. Derive the optimal action in terms of the MDP function $p(s', r|s, a)$, policy π , if any, as required, and other quantities that are said to be known to the agent.

The action a_t must maximize

$$E[R_{t+1} + \gamma V_*(S_{t+1}) | S_0=s_0, S_1=s_1, \dots, S_{t-1}=s_{t-1}, A_t=a]$$

$$= \sum_{s'} \sum_{\pi} (\pi + \gamma V_*(s')) p(s', \pi | s_0, s_1, \dots, s_{t-1}, a)$$

$$= \sum_{s'} \sum_{\pi} \sum_s (\pi + \gamma V_*(s')) p(s', \pi, s | s_0, s_1, \dots, s_{t-1}, a)$$

$$= \sum_{s'} \sum_{\pi} \sum_s (\pi + \gamma V_*(s)) p(s', \pi | s_0, s_1, \dots, s_{t-1}, a) p(s | s_0, s_1, \dots, s_{t-1})$$

$$= \sum_{s'} \sum_{\pi} \sum_s (\pi + \gamma V_*(s)) p(s', \pi | s, a) p(s | s_{t-1})$$

$$a_t = \operatorname{argmax}_{a \in A} \sum_{s'} \sum_{\pi} \sum_s (\pi + \gamma V_*(s)) p(s', \pi | s, a) p(s | s_{t-1})$$

One could further expand:

$$p(s | s_{t-1}) = \sum_{a_{t-1}} p(s, a_{t-1} | s_{t-1})$$

$$= \sum_{a_{t-1}} p(s | s_{t-1}, a_{t-1}) p(a_{t-1} | s_{t-1})$$

$$= p(s | s_{t-1}, \operatorname{argmax}_a E[R_t + \gamma V_*(S_t) | s_{t-1}, a_{t-1}])$$

Clearly stating that the expectation conditioned (as shown) must be maximized

Up to 9/15

We didn't observe the state S_t . We average over all values s that S_t may take.

Up to 10/15

Expansion using MDP V_* , find expression to calculate action a_t .

12/15

3/15

Question 5. 15 marks You are given two value functions $v_1(s)$ and $v_2(s)$, where state $s \in \{1, 2, 3, 4, 5, 6\}$. Let $v_1(s) = v_*(s)$ for $s = 1, 2, 3$ and let $v_1(s) = v_*(s) - \epsilon s^2$ for $s = 4, 5, 6$. Here $v_*(s)$ is the optimal value and $\epsilon > 0$. Further, let $v_2(s) = v_*(s)$ for $s = 4, 5, 6$ and let $v_2(s) = v_*(s) - \epsilon s^3$ for $s = 1, 2, 3$. Derive the optimal policy function for any MDP $p(s', r|s, a)$.

For every s , $V_{\pi}(s) = \max(V_1(s), V_2(s))$.

Optimal policy

$$\pi_*(s) = \underset{a \in A(s)}{\operatorname{argmax}}. \sum_{s'} \sum_{r} (r + \gamma \max(V_1(s'), V_2(s'))) p(s', r|s, a).$$

15/15